RE07 Global Infrastructure Investment and Management

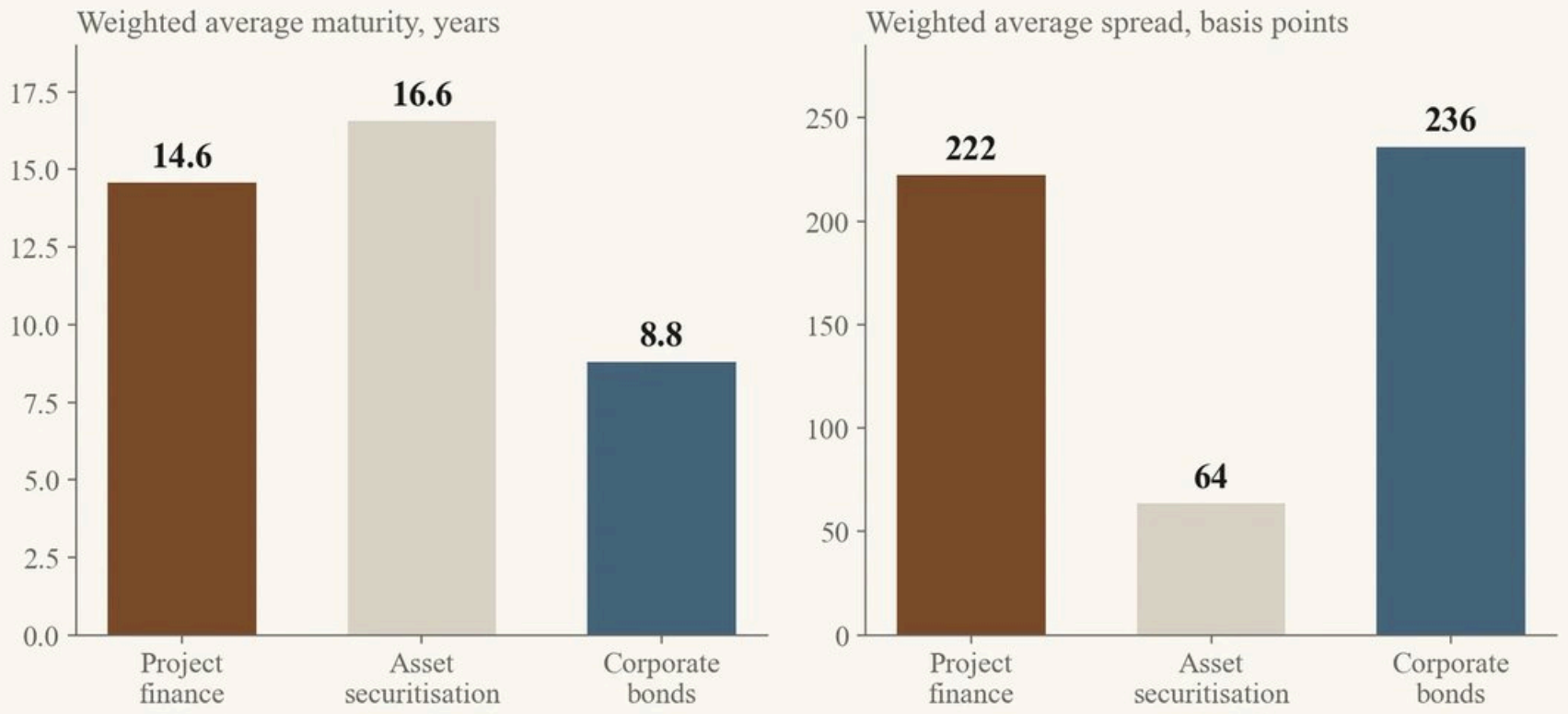
4. Infrastructure Financing and Deal Structuring

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Lecture Outline

Project finance and corporate borrowing
Default, recovery and risk allocation
Procurement and the cost of competition
Guarantees and credit enhancement
Three models for sizing, pricing and backing project debt
Case Study: Topic to Be Agreed (Steven Norris, Chairman, Soho Estates)

Project Finance and Corporate Bonds



Source: Pinto and Santos (2019), working paper of 18 April 2019, Table 3 (page 36) and pages 4, 16 and 29. Deals closed by borrowers in 19 European countries.

The sample: 583 project finance and 3,949 corporate bond deals closed by European borrowers, 2000 to 2016

Tenor: project finance runs 14.6 years on average, corporate bonds 8.8

Price: mean spreads of 222 and 236 basis points are not significantly different

The choice: made when information, agency problems and issuance costs make a plain bond dear, and the borrower is less creditworthy

Hyperion: The Lease as the Contract



The vehicle: a joint venture, 80 per cent Blue Owl, 20 per cent Meta, with USD 27 billion of debt on about USD 30 billion of assets

The revenue: five four-year leases from 2029 to the bond's maturity in 2049; Meta may end each

The guarantee: if Meta leaves, it covers any shortfall between the sale and a minimum value

The accounting: neither the lease nor the guarantee is on Meta's balance sheet until it takes effect

Van Nieuwerburgh (2026), NBER Working Paper 35865, pages 8 to 11 and Figure 2. Timeline drawn from the five four-year leases starting in 2029; the renewal dates are my arithmetic.

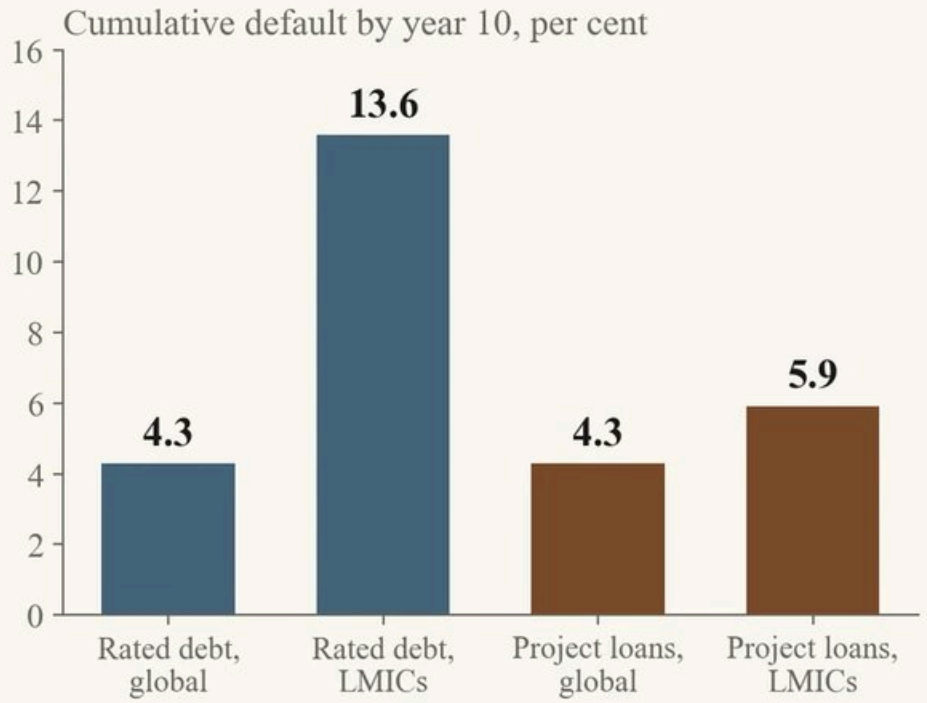
Hyperion: Project Debt against Corporate Debt

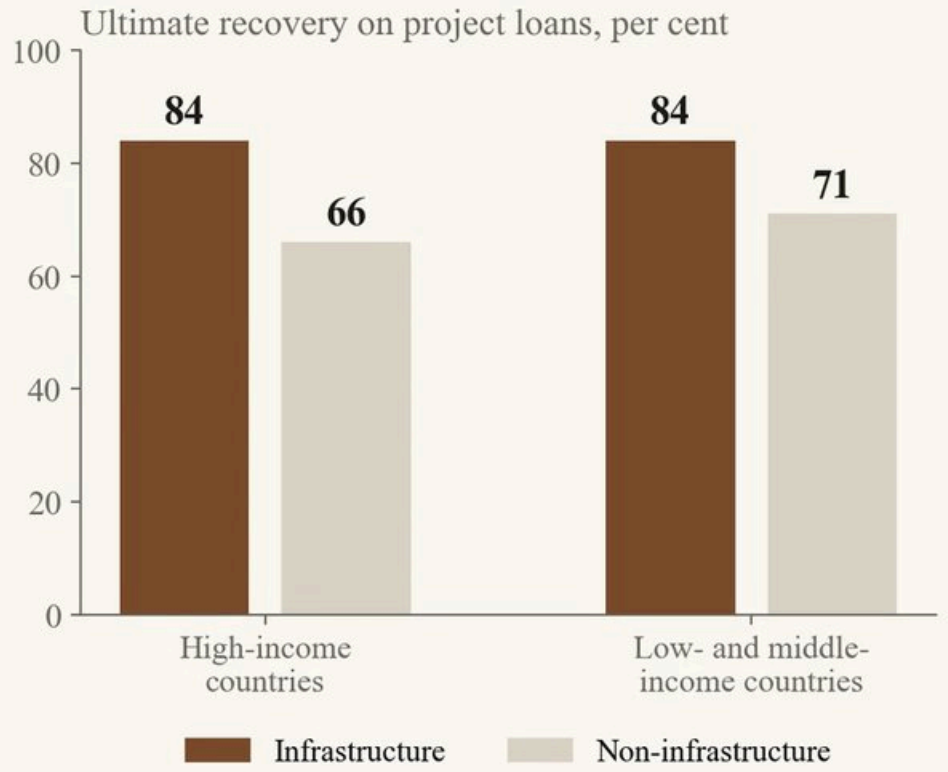
	Meta borrowing for itself	Beignet project debt
What backs the debt	Meta's balance sheet and cash flow	Lease payments and the residual value guarantee, through bankruptcy-remote vehicles
Rating	Meta's own corporate rating	A+, one notch below Meta's
Leverage	About 25 per cent against book equity; 4 per cent against market equity	About 90 per cent of asset value, with cover of 1.12
Yield	At least 100 basis points lower, on the paper's estimate	6.58 per cent
Extra interest over the life	None	More than USD 5 billion
What Meta gets	Assets and debt on its own balance sheet	An asset-light profile and lower reported leverage

Five-year credit default swap spreads on Meta widened from 56.1 to 80.2 basis points between 21 November 2025 and 22 July 2026, which the paper reads as the market pricing opacity.

Van Nieuwerburgh (2026), NBER Working Paper 35865, pages 8 to 11. Meta's own leverage is from note 4 and the yield comparison from page 10; the 100 basis points is the author's estimate.

Default and Recovery in Infrastructure Debt





Source: World Bank Group and Global Infrastructure Hub, Infrastructure Monitor 2024, financial performance chapter, pages 18 and 22 (Moody's data). LMICs are low- and middle-income countries.

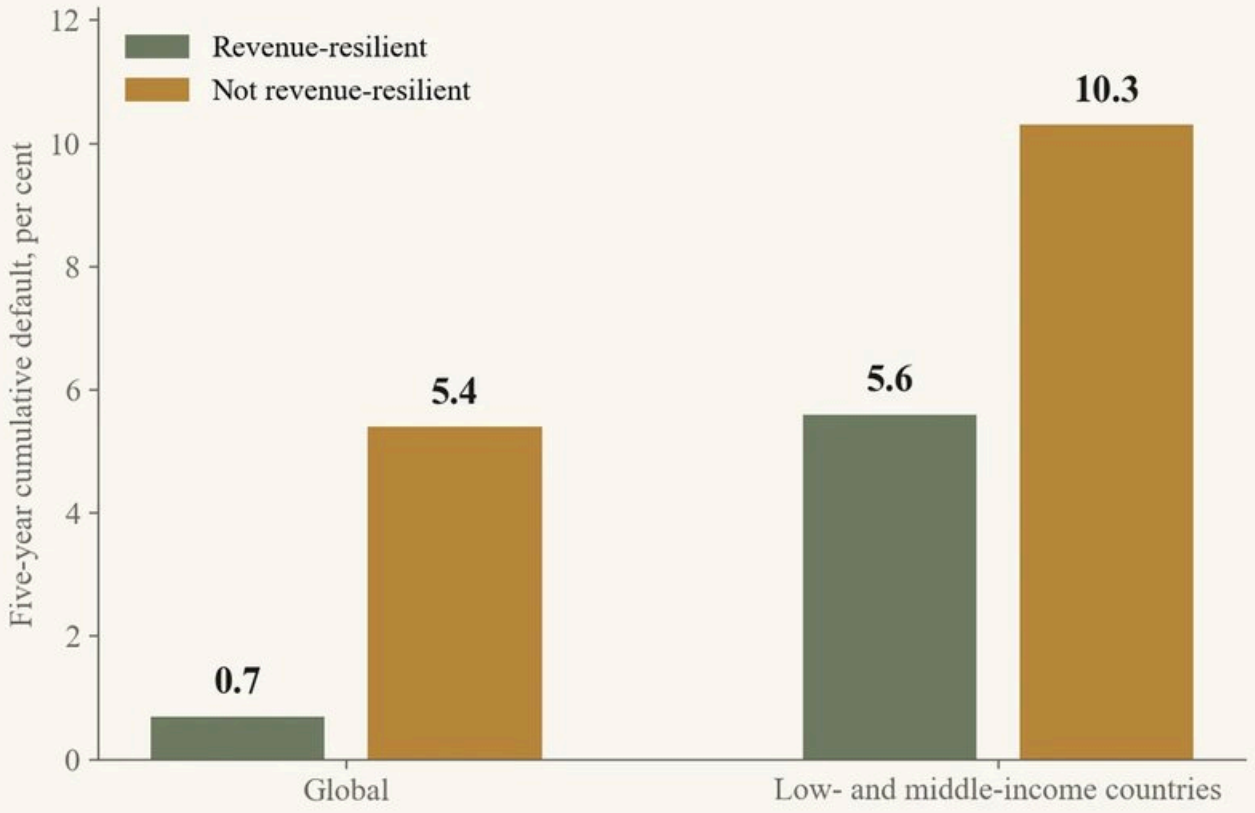
Rated debt: 4.3 per cent cumulative default by year 10 globally, 13.6 per cent in low- and middle-income countries

Project loans: 4.3 per cent globally and 5.9 per cent in low- and middle-income countries

Grade: investment-grade infrastructure debt defaulted below 2 per cent over ten years

Recovery: 84 per cent ultimate recovery on infrastructure project loans against 66 per cent for others in high-income countries

Revenue Resilience and Default



Definition: availability-based PPPs and regulated utilities in electricity, gas and water

The gap: five-year default of 0.7 per cent against 5.4 globally, 5.6 against 10.3 in low- and middle-income countries

Availability payments: a ten-year default rate of 0.6 per cent, but recovery lower where the state is the only payer

Design: contracts, regulation and payment guarantees can create revenue resilience

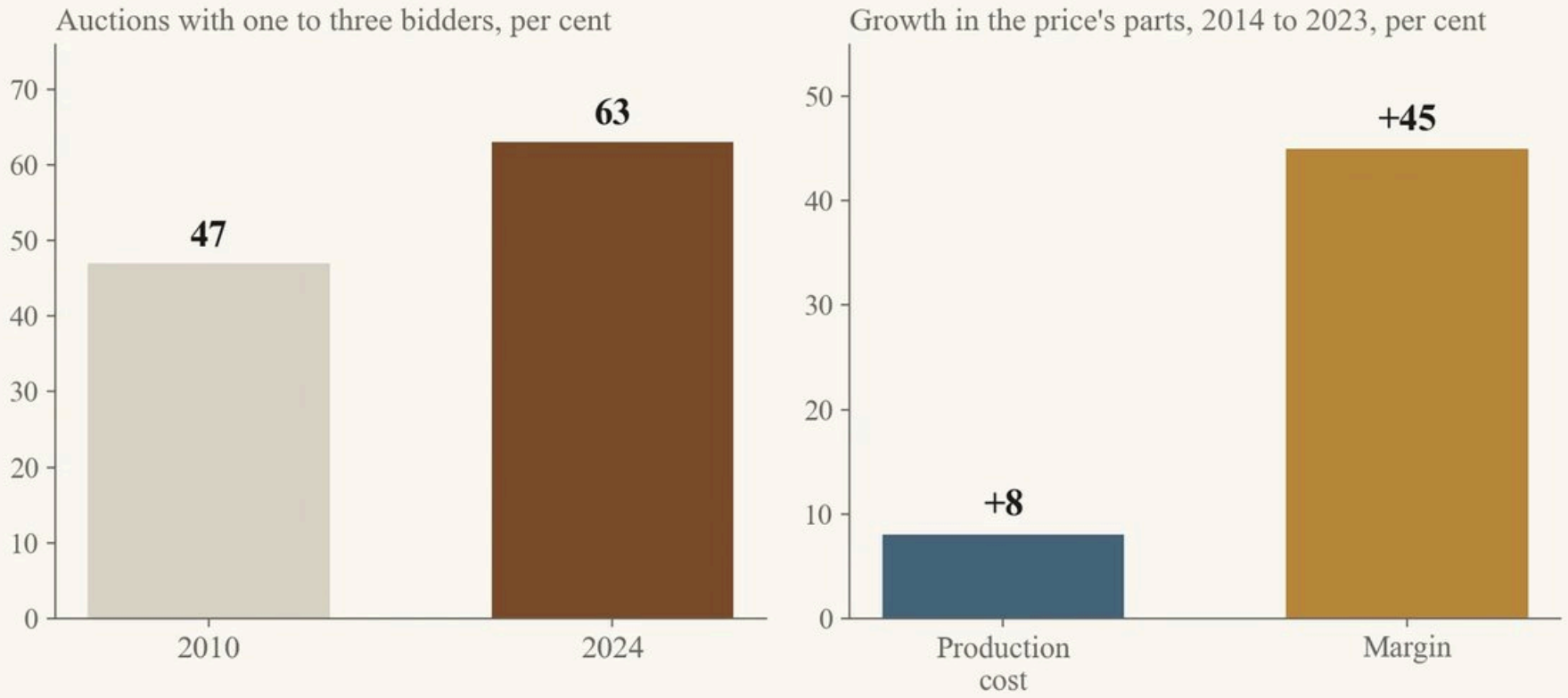
World Bank Group and Global Infrastructure Hub, Infrastructure Monitor 2024, financial performance chapter, pages 19 and 20 (Moody's data). Rated infrastructure debt originated 1995 to 2023.

Risk Allocation Matrices

Risk	Who bears it	Qualification in the matrix
Land and site	The contracting authority	Shared where the private partner's design affects the site
Design	The private partner	The authority shares it if the specification is too prescriptive
Construction	The private partner	Unless the work depends on infrastructure the authority must complete
Demand, developed markets	Increasingly the authority	Earlier projects put it on the private partner; forecasts fell short and some firms failed
Demand, emerging markets	The private partner, by default	A subsidy or a cap and collar may be needed if demand falls below a level

Global Infrastructure Hub (2016), Allocating Risks in PPP Contracts, 2016 edition, Risk Matrix 1, toll road (DBFO), pages 16, 19 and 24. The allocations are stated from the matrix's rationale text.

Thin Competition in Highway Auctions



Source: Currier (2026), Competition and the Cost of U.S. Infrastructure, September 2026 draft, pages 1 to 6. State highway auctions, 2002 to 2024; the decomposition covers 2014 to 2023.

The data: 1.3 million bids in state highway auctions, 2002 to 2024

Competition: the median auction has three bidders; one or two bidders in about a third of projects

The effect: one more bidder lowers the winning bid by about 10 per cent

The price: of the real rise in price from 2014 to 2023, margins grew 45 per cent and production costs 8 per cent

Auctions and Renegotiation

Study	Setting	Finding
Bajari, McMillan and Tadelis (2003)	Private building contracts, Northern California, 1995 to 2000	44 per cent negotiated against 18 per cent by open bidding; auctions do badly when projects are complex, contracts incomplete and bidders few
Decarolis (2013)	Italian public works, first-price auctions introduced at different dates	At least half of the savings from lower winning bids are lost to renegotiation; screening the lowest bid cuts the savings by a third
Makovšek and Bridge (2021)	Review of evidence on procurement choices	Evidence does not match theory for larger projects; incentives for certainty carry a cost premium even without renegotiation

Bajari, McMillan and Tadelis (2003), NBER Working Paper 9757, page 1; Decarolis (2013), abstract; Makovšek and Bridge (2021), page 278.

Guarantees and Credit Enhancement

Partial credit guarantee: covers lenders for a specified portion of the debt, whatever the cause of default

Partial risk guarantee: covers named events such as government non-payment; multilateral banks prefer partial cover to limit moral hazard

Scale in 2017: on private estimates, 5 per cent of multilateral bank operations and 45 per cent of private resources mobilised

Effect: resilient revenue, which payment guarantees can support, halves default in low- and middle-income countries

A private version: Meta's guarantee of a floor on the value of the Hyperion assets

Pereira dos Santos and Kearney (2018), IDB Technical Note 1358, pages 4, 8, 26 and 31 (a 2018 note; shares from private estimates); World Bank Group and Global Infrastructure Hub, Infrastructure Monitor 2024, page 19.

Three Models for Financing a Project

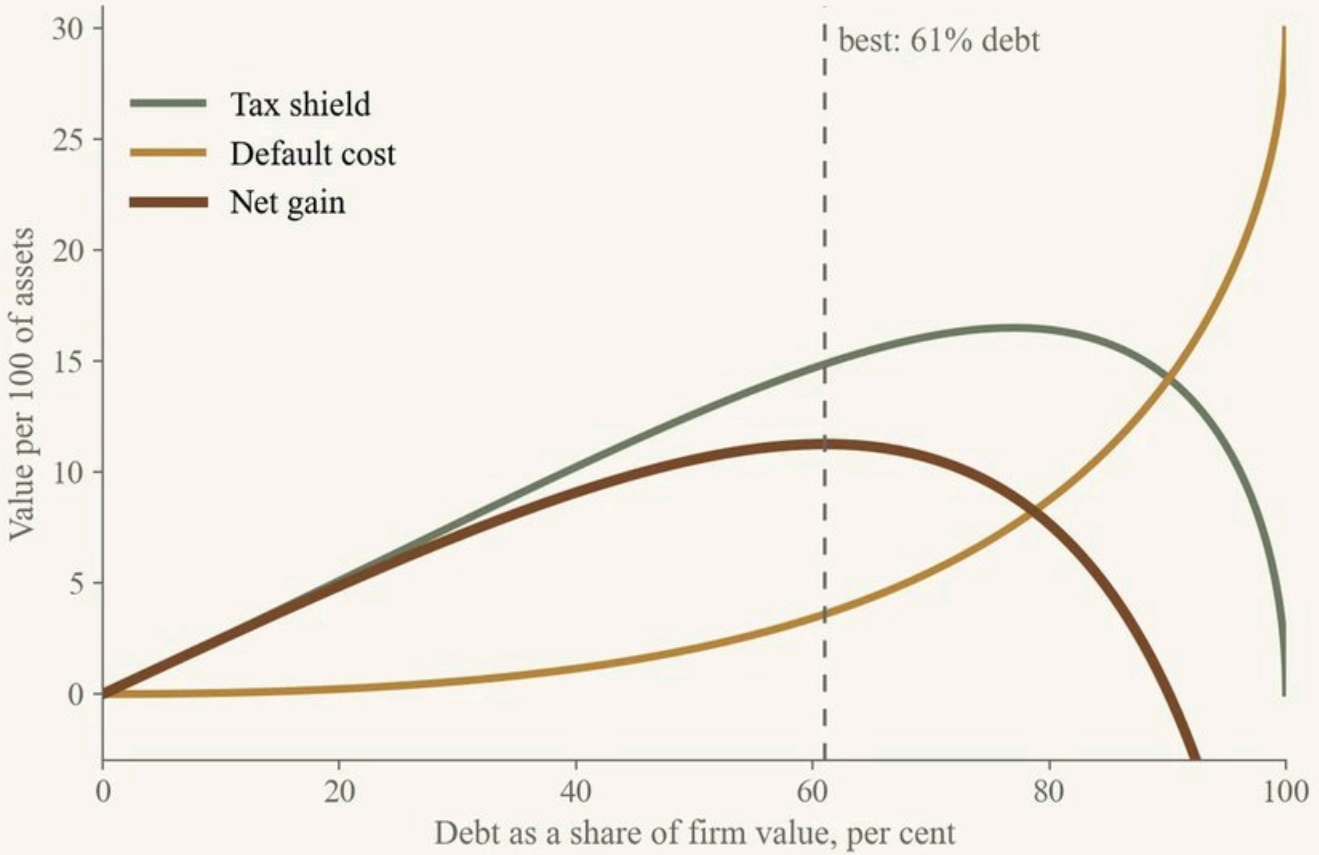
Leland (1994): how much debt: the tax shield against the cost of default

EDHEC Infrastructure Institute (2016): how much margin: debt service cover as a persistent random process

Merton (1977): what a promise to pay the lender is worth: a guarantee as an option

How they fit: a loan has a size, a margin and a backstop, and each model tests one of them

Leland (1994): Tax Shields and Default Costs



The gain: interest is deducted from tax; at the best coupon the shield is worth 14.9 per 100 of assets

The cost: default loses 30 per cent of assets; at the best coupon its weight is 3.6

The choice: equity holders stop paying when the asset falls to 34, against a value of 100

The best: a coupon of 3.70 a year, debt of 61 per cent of firm value, and firm value 11.3 per cent higher

Leland (1994), Journal of Finance 49(4), the model; closed forms derived and checked (models_w4.py). Illustrative inputs: volatility 15 per cent, payout 6, risk-free rate 4, tax 25, loss in default 30 per cent.

The Optimal Leverage of a Low-Risk Asset



Risk: volatility of 8, 15 and 30 per cent gives best debt shares of 65, 61 and 55 per cent, and spreads of 91, 144 and 317 basis points

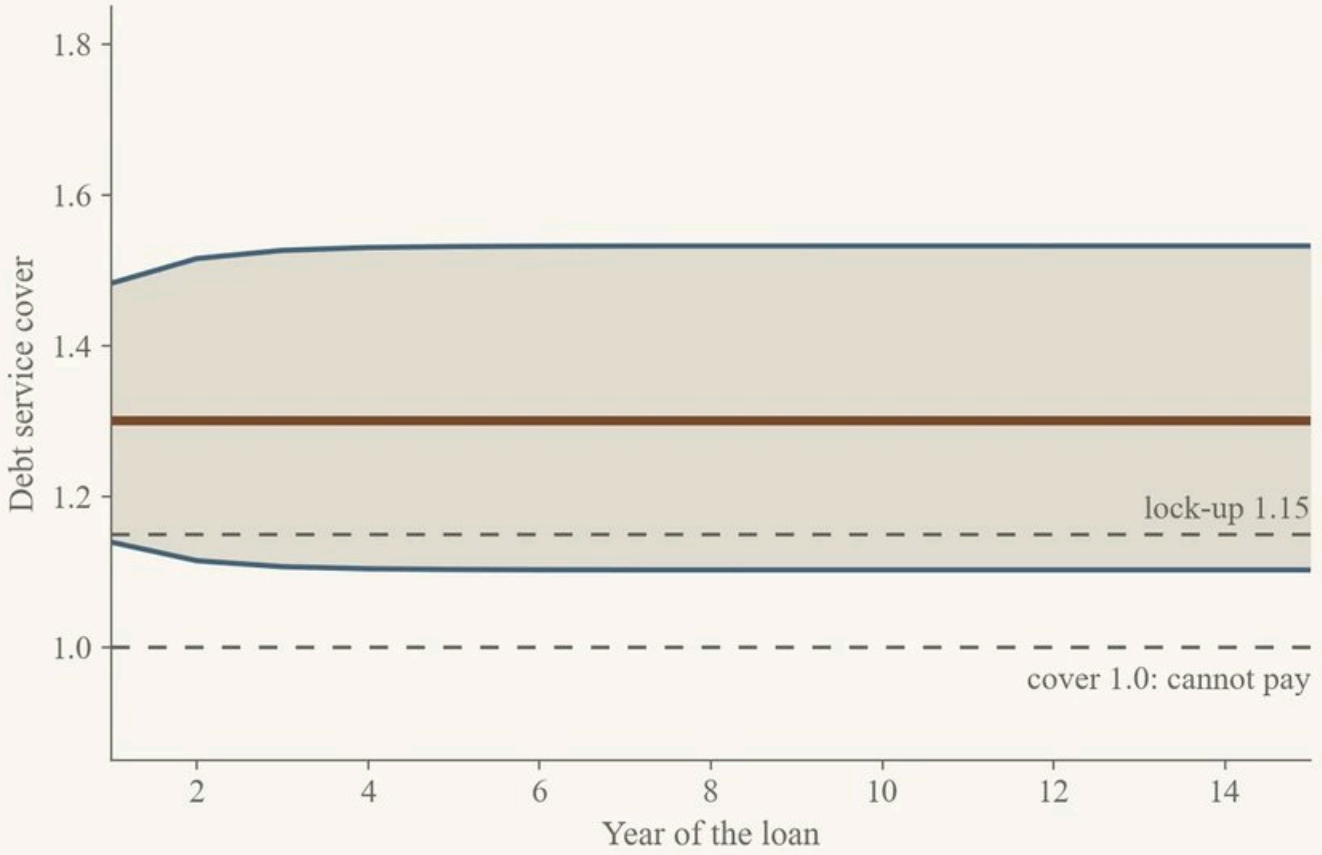
The cost of failure: with 10 per cent of assets lost in default rather than 30, the share at 15 per cent volatility is 72 per cent, not 61

Project finance: a single-asset vehicle with contracted revenue and step-in rights tries to lower both

The evidence: senior leverage of around 90 per cent for projects with little market risk, above the page's generic figure

Author's model (models_w4.py), after Leland (1994); illustrative inputs. EDHEC Infrastructure Institute-Singapore (2016), pages 8 and 9, for the practice of around 90 per cent senior leverage.

Debt Service Cover and the Lender's Sizing Rule



Cover: cash flow available for debt service over the instalment; below 1.0 the project cannot pay

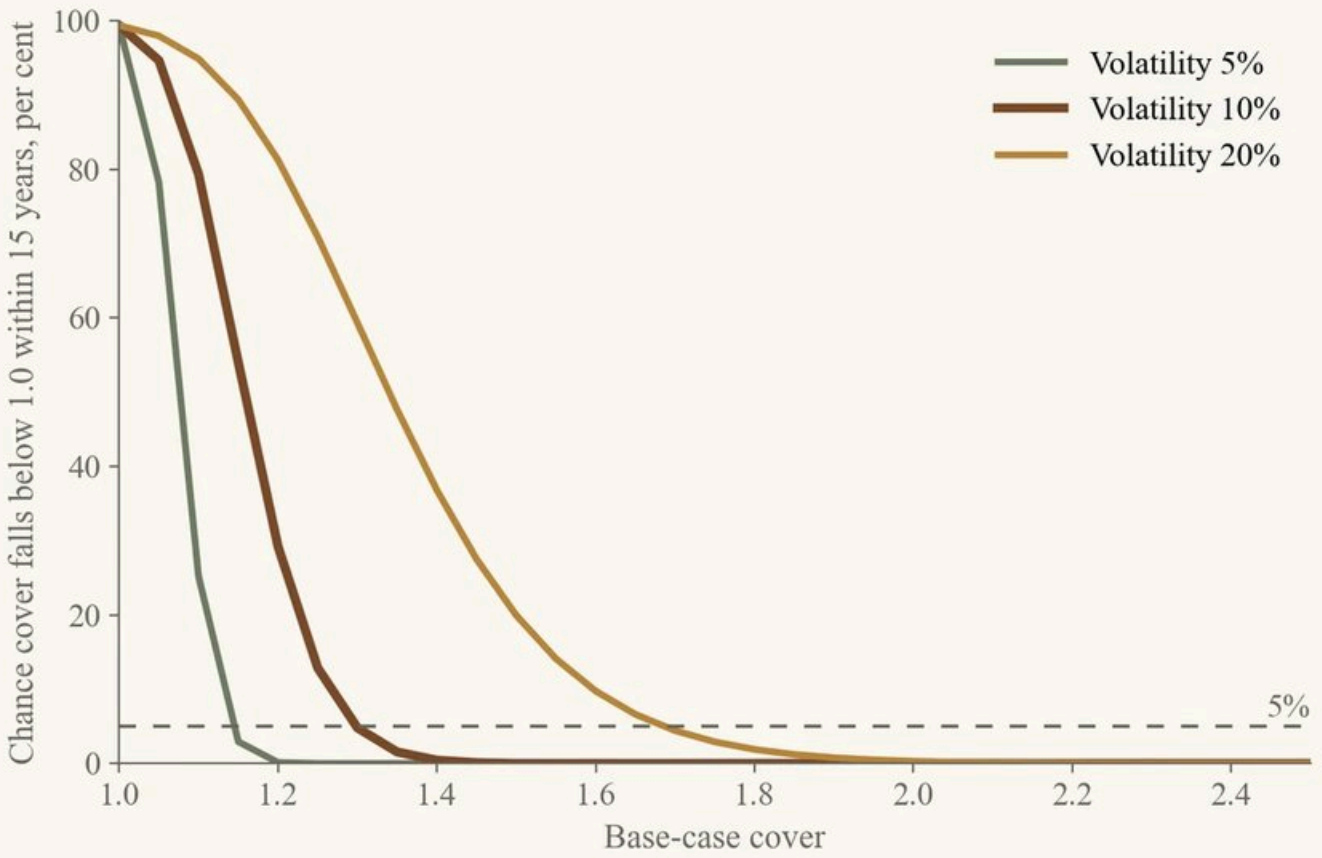
The evidence: more than 200 projects over 15 years: cover is close to lognormal, persistent, and less volatile where revenue is contracted

The model: at cover 1.30, volatility 10 per cent and persistence 0.6, the fifth percentile falls to 1.10 by year 15

The loan: level instalments at that cover support debt of 747 per 100 of annual cash flow, at 6 per cent over 15 years

EDHEC Infrastructure Institute-Singapore (2016), Cash Flow Dynamics of Private Infrastructure Project Debt, pages 6, 15, 32 and 37. Author's model (models_w4.py); illustrative inputs.

Sizing Debt against the Probability of Breach



Breach: at cover 1.30 and 10 per cent volatility, a 4.8 per cent chance of falling below 1.0 in 15 years

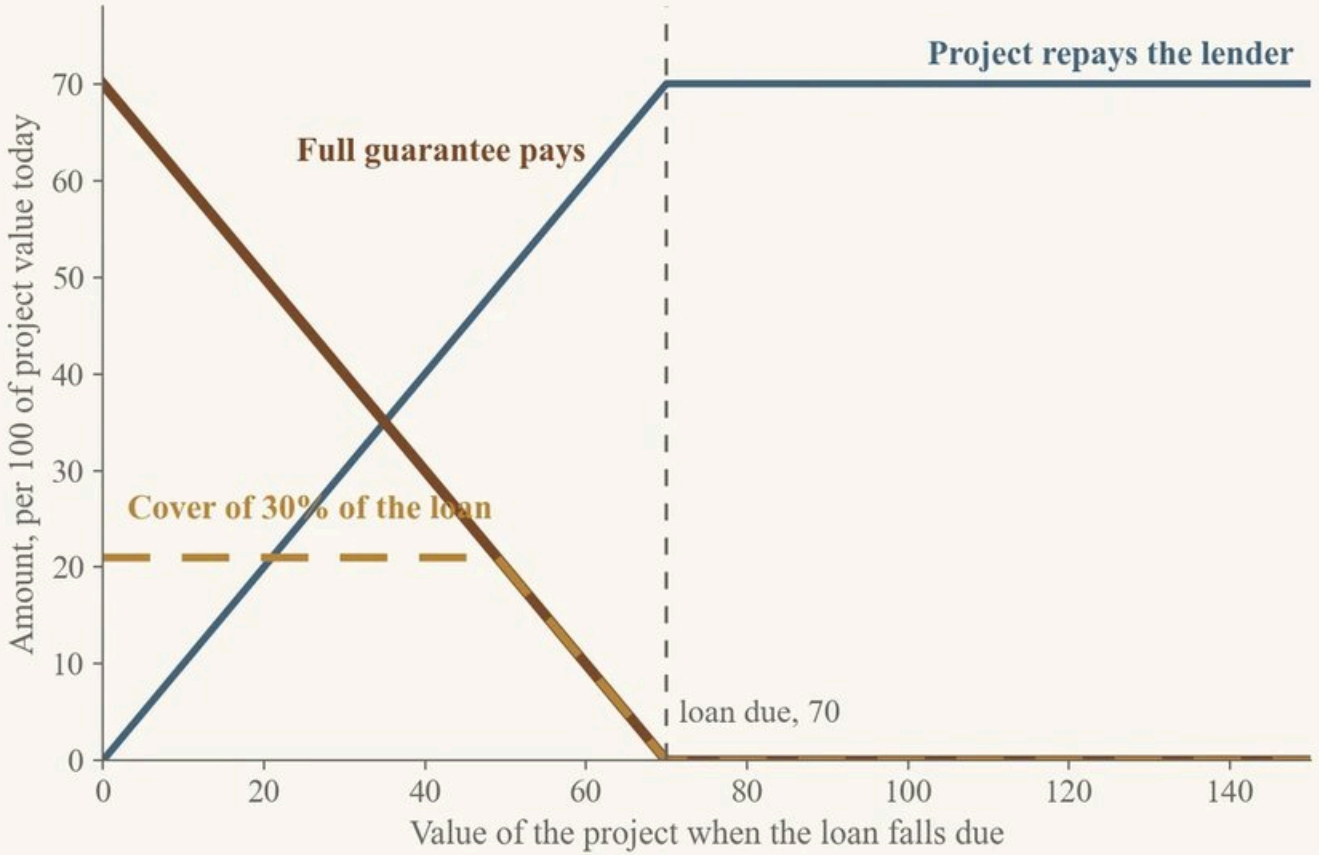
Margin: for a 5 per cent tolerance the lender needs 1.14, 1.30 and 1.69 at volatilities of 5, 10 and 20 per cent

Debt: those covers support 852, 748 and 576 per 100 of annual cash flow

Hyperion: cover of 1.12 over 20 years is consistent with a 5 per cent breach chance only if volatility is about 4 per cent

Author's model (models_w4.py); illustrative inputs: 15 years, persistence 0.6. EDHEC Infrastructure Institute-Singapore (2016) for the cover ratio and its dynamics; Van Nieuwerburgh (2026), page 9, for cover of 1.12.

Merton (1977): A Guarantee Is a Put Option



The payoff: the guarantor pays the shortfall between the loan and the project's value, up to the share covered

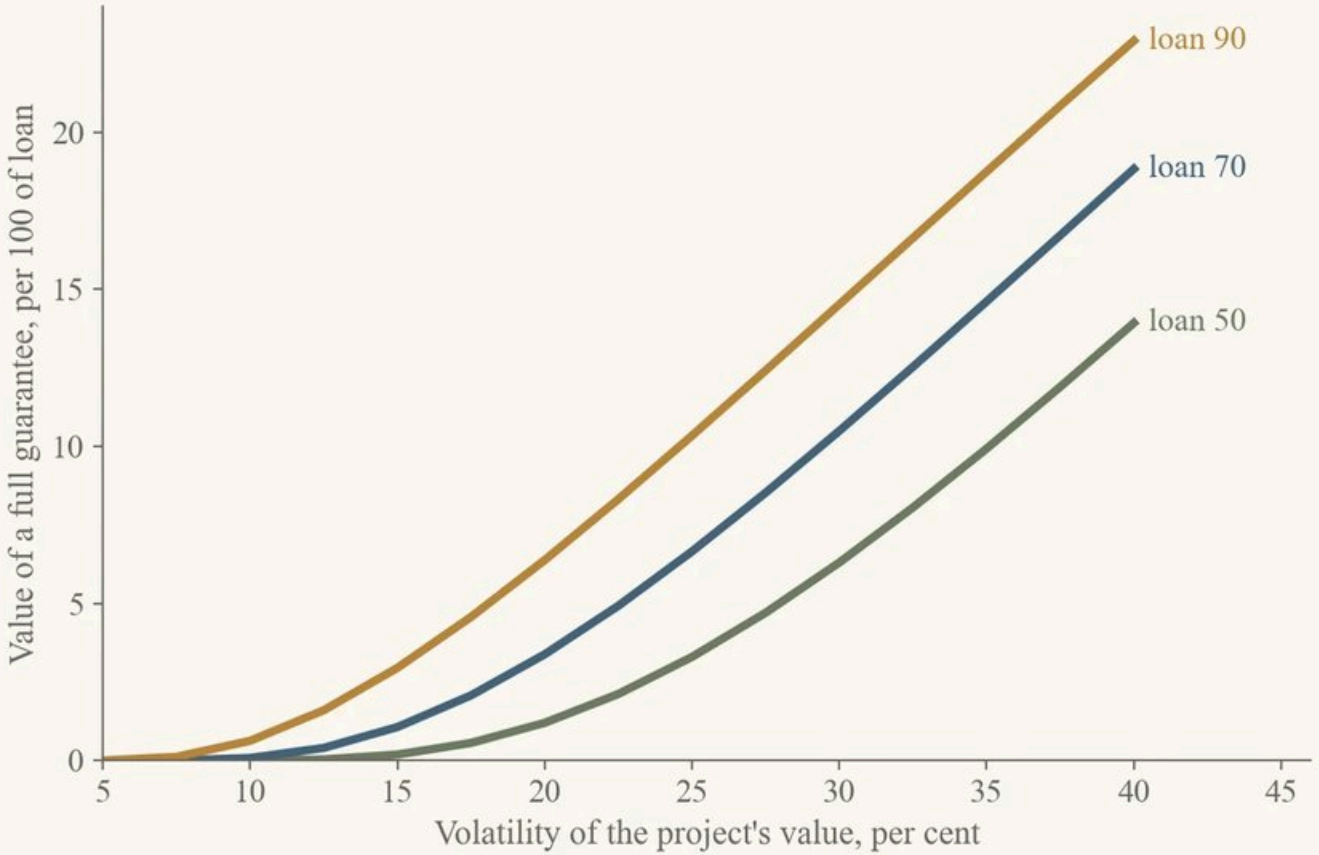
The price: a put option on the project struck at the loan; a partial guarantee is a put spread

Two prices: the market value uses the risk-free rate; the budget's expected cost uses expected growth, and the gap is the risk premium

The lender: the guaranteed loan is riskless, so its interest saving equals the guarantee's price

Merton (1977), Journal of Banking and Finance 1(1), 3 to 11 (not retrieved; formulas derived and checked); author's model (models_w4.py). Illustrative: loan 70, project value 100, ten years.

The Value of a Loan Guarantee



Debt: a loan of 50, 70 and 90 per 100 of value has a full guarantee worth 0.19, 1.07 and 2.95 per 100 of loan

Risk: at a loan of 70, volatility of 8 and 30 per cent gives 0.01 and 10.49

The budget: at a loan of 90 the expected cost is 0.53, against a market value of 2.95

Partial cover: the first 30 per cent of a loan of 90 keeps 84 per cent of the full guarantee's value

Author's model (models_w4.py), after Merton (1977); illustrative inputs: project value 100, ten years, risk-free rate 4 per cent, real-world growth 8 per cent.

Three Models Compared

Model	Explains	Prediction	Evidence and limit
Leland (1994)	How much debt	The best debt share falls with risk and with the cost of failure	Around 90 per cent senior leverage where market risk is low; one asset, permanent debt
EDHEC (2016) on cover	How much margin	The cover a lender needs rises with the volatility of cash flow	More than 200 projects; pooled across projects and years; a breach is not a default
Merton (1977)	What a guarantee is worth	Value rises with loan to value and risk, and exceeds expected cost	Default and recovery data show the tail; one payment, tradable asset

Leland (1994); EDHEC Infrastructure Institute-Singapore (2016), pages 8 and 9; Merton (1977); World Bank Group and Global Infrastructure Hub, Infrastructure Monitor 2024, pages 18 and 22.

Case Study: Topic to Be Agreed (Steven Norris, Chairman, Soho Estates)

The project and its sponsor: what was built, by whom, and why

The contracts and the risks they allocate: who bore which risk

The financing and who lends: debt, equity, tenor, margin

The public support: guarantees, subsidy, indemnities

Discussion: questions for the guest

Summary

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